

Requirements Specification – Team 05

Functional Requirements

R-TRACK-01

Requirement: The robot shall maintain lateral tracking error within safe tolerance during S1 nominal scenario.

Acceptance Criterion: $|e_{\text{line}}| < 0.15 \text{ m}$ for all $t \in [0, 75] \text{ s}$

Justification: Ensures predictable and stable motion behaviour during nominal operation.

R-TRACK-02

Requirement: The robot shall minimize cumulative tracking error over the full S1 scenario.

Acceptance Criterion: $\text{IAE} < 0.030 \text{ m*s}$

Justification: Performance optimization requirement for smooth and accurate line following.

R-OBS-01

Requirement: The robot shall avoid collision with static obstacles during S2 scenario.

Acceptance Criterion: Number of obstacle contacts = 0

Justification: Safety requirement to prevent hardware damage and ensure operational reliability.

R-SAFE-01

Requirement: The controller shall limit velocity during sensor fault condition.

Acceptance Criterion: $v \leq 0.45 \text{ m/s}$ when $\text{fault_flag} = 1$

Justification: Safe-state behavior requirement during abnormal conditions.

R-CTRL-01

Requirement: The controller shall saturate motor commands within actuator limits.

Acceptance Criterion: $-1.0 \leq u_L \leq 1.0$ and $-1.0 \leq u_R \leq 1.0$ at all times

Justification: Protects actuators and ensures commands remain within physical limits.

Non-Functional Requirements

R-PERF-01

Requirement: The robot shall complete S1 scenario within the specified time limit.

Acceptance Criterion: $t_{\text{final}} \leq 75 \text{ s}$

Justification: Efficiency requirement for acceptable system performance.

R-ENERGY-01

Requirement: The controller shall maintain acceptable energy usage during S1 scenario.

Acceptance Criterion: Energy proxy $< 65 \left(\int (|u_L| + |u_R|) dt \right)$

Justification: Sustainability consideration based on life-cycle assessment screening.

R-USAB-01

Requirement: The HMI shall display key system variables clearly to the user.

Acceptance Criterion: e_{line} and v displayed at $\geq 10 \text{ Hz}$ refresh rate

Justification: Ensures usability and operator awareness.